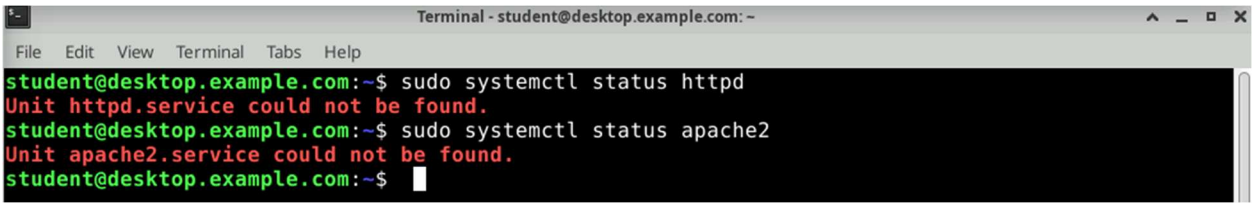
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Apache Server

Installation

The purpose of this assignment was to install and configure the Apache HTTP Server on a Linux system and create a basic “Hello World” webpage. I used the **Virginia Cyber Range (VCR)** Ubuntu environment, since this is the same platform used during class demonstrations.

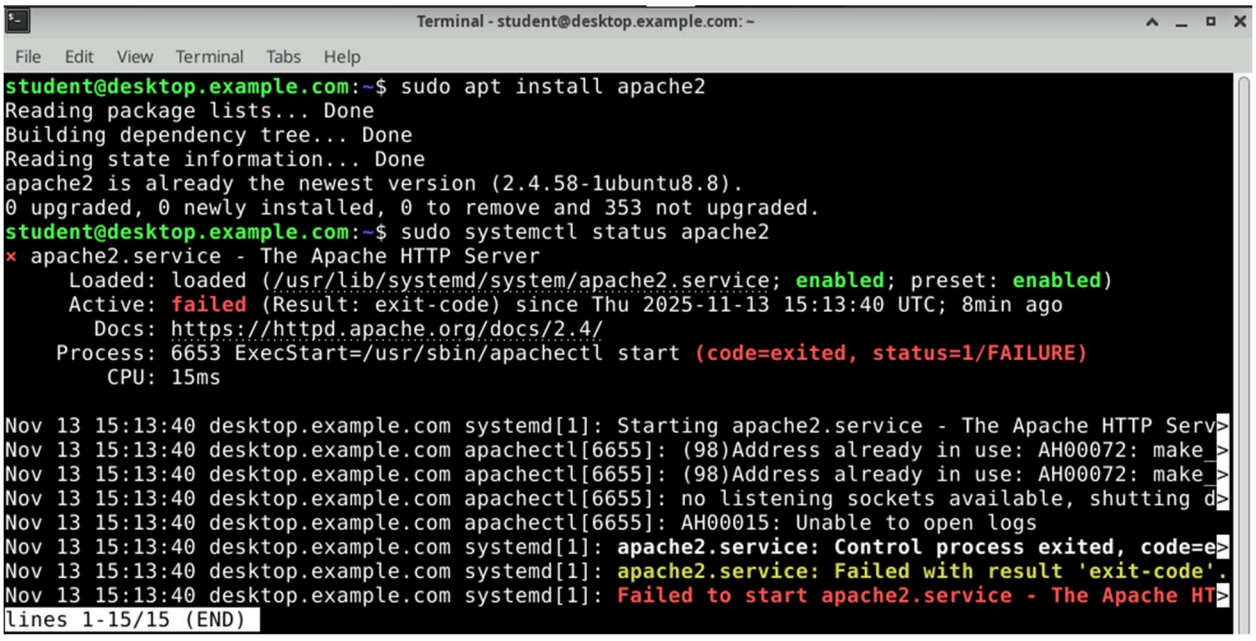
Before installation, I verified whether Apache was already present in the environment by running: sudo systemctl status httpd & sudo systemctl status apache2



This confirmed that Apache was not installed, and I proceeded with a full installation.

First, I updated the package index: sudo apt update
Then I installed Apache: sudo apt install apache2
After installation, I checked the service: sudo systemctl status apache2
The service showed: loaded, but failed (status=1/FAILURE)

This indicated that Apache was installed successfully but could **not** start due to an underlying issue. That led me into the troubleshooting phase of my assignment.



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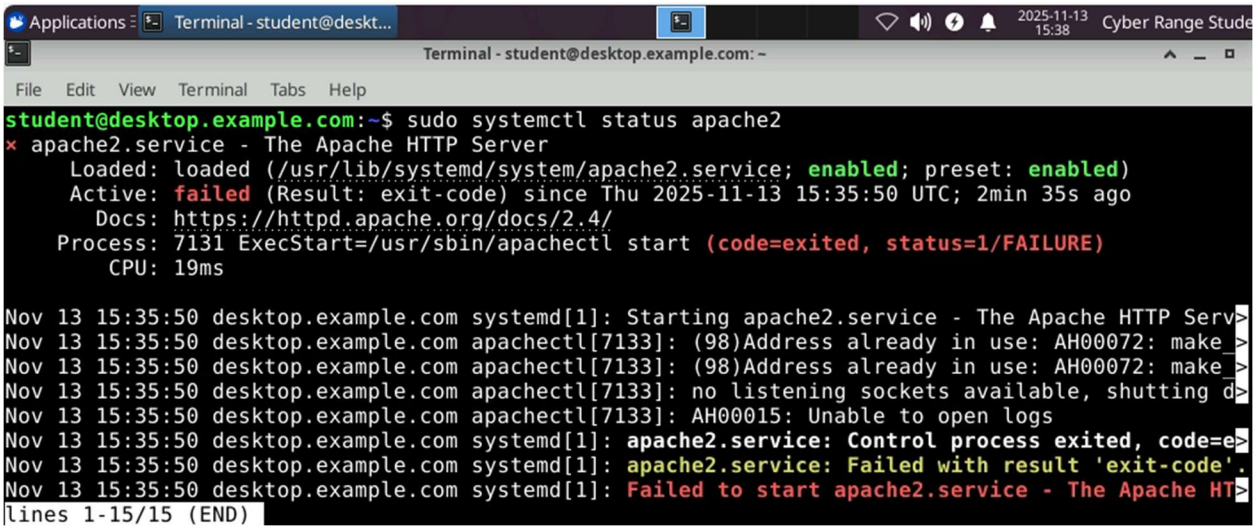
Troubleshooting & Configuration

Before troubleshooting ports, I confirmed ownership and permissions of /var/www/html:

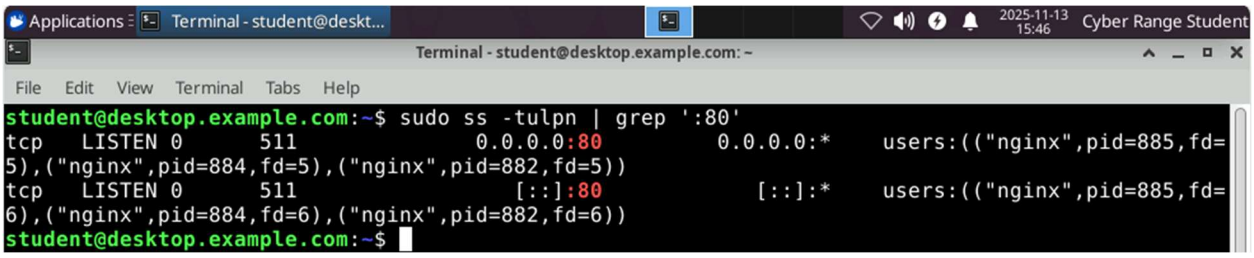
```
student@desktop.example.com:~$ ls -l /var/www/html
total 12
-rw-r--r-- 1 root root 10671 Nov 13 15:13 index.html
student@desktop.example.com:~$ sudo chmod 750 /var/www/html
student@desktop.example.com:~$ sudo chown student:www-data /var/www/html
student@desktop.example.com:~$
```

```
student@desktop.example.com:~$ ls -ld /var/www/html
drwxr-x--- 2 student www-data 4096 Nov 13 15:13 /var/www/html
```

Everything looked normal, so file permissions were not the cause of the failure.



I had to do a little bit of research, and I found a command that can help me discover if the issue that I'm having is a port conflict. sudo ss -tulpn | grep ':80'. And of course, it returned that nginx is using port :80. Since I am using the Virginia Cyber Range and we have already installed and removed apache2 and nginx, it looks like apache is not able to take over the port just yet. Let's fix that.

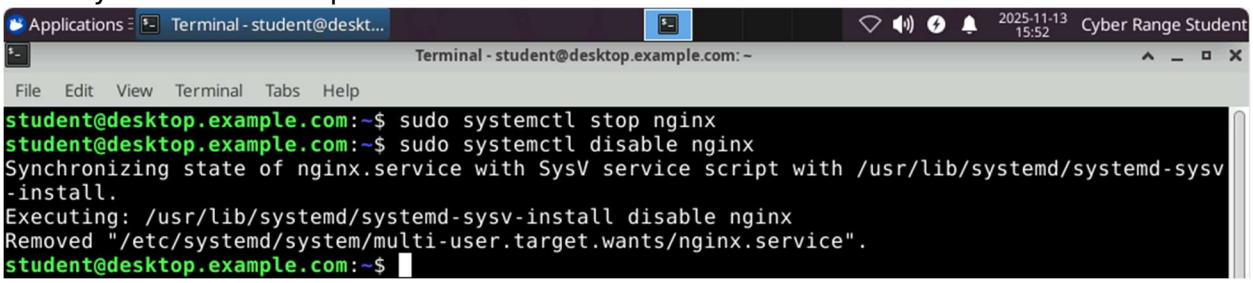


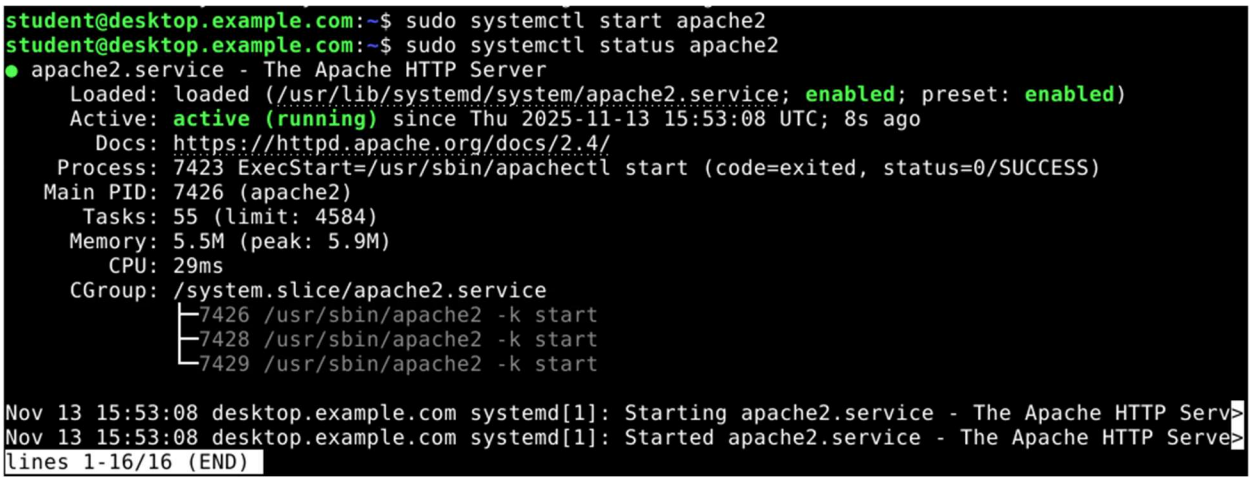
First, we're going to stop nginx and disable nginx and restart apache2

sudo systemctl stop nginx
sudo systemctl disable nginx

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sudo systemctl restart apache2
sudo systemctl status apache2





Success! We can proceed to the next step.

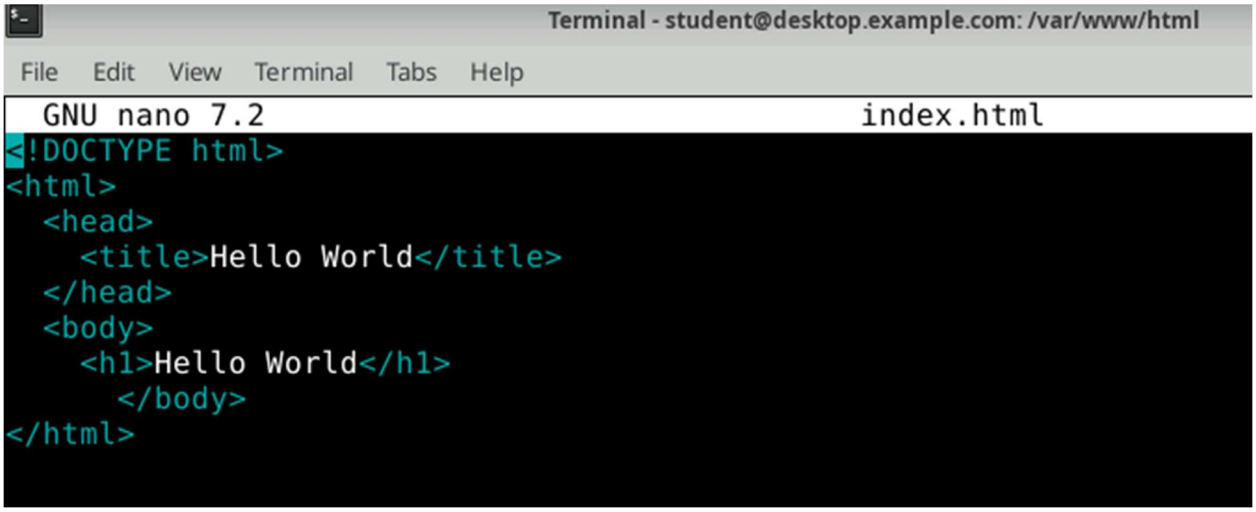
Website Setup

Inside the Apache document root:

cd /var/www/html

I created and edited the main HTML file:

sudo nano index.html

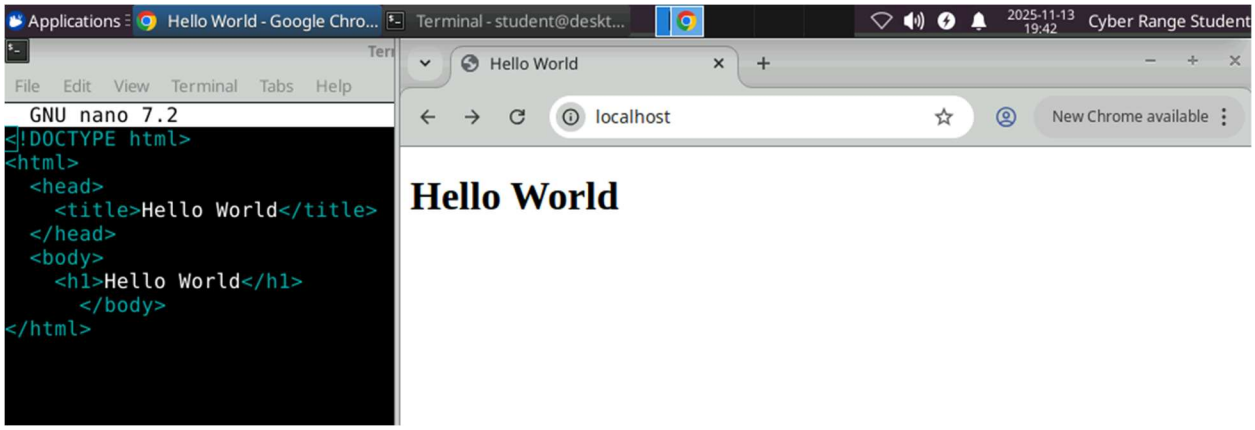


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Initially, nginx cached an older index.html file, so Apache wasn't showing my updated page. I solved this by:

1. Renaming the file
2. Renaming it back to index.html

After that, refreshing the browser correctly loaded the **Hello World** page.



We can check to see any other security considerations like a firewall.

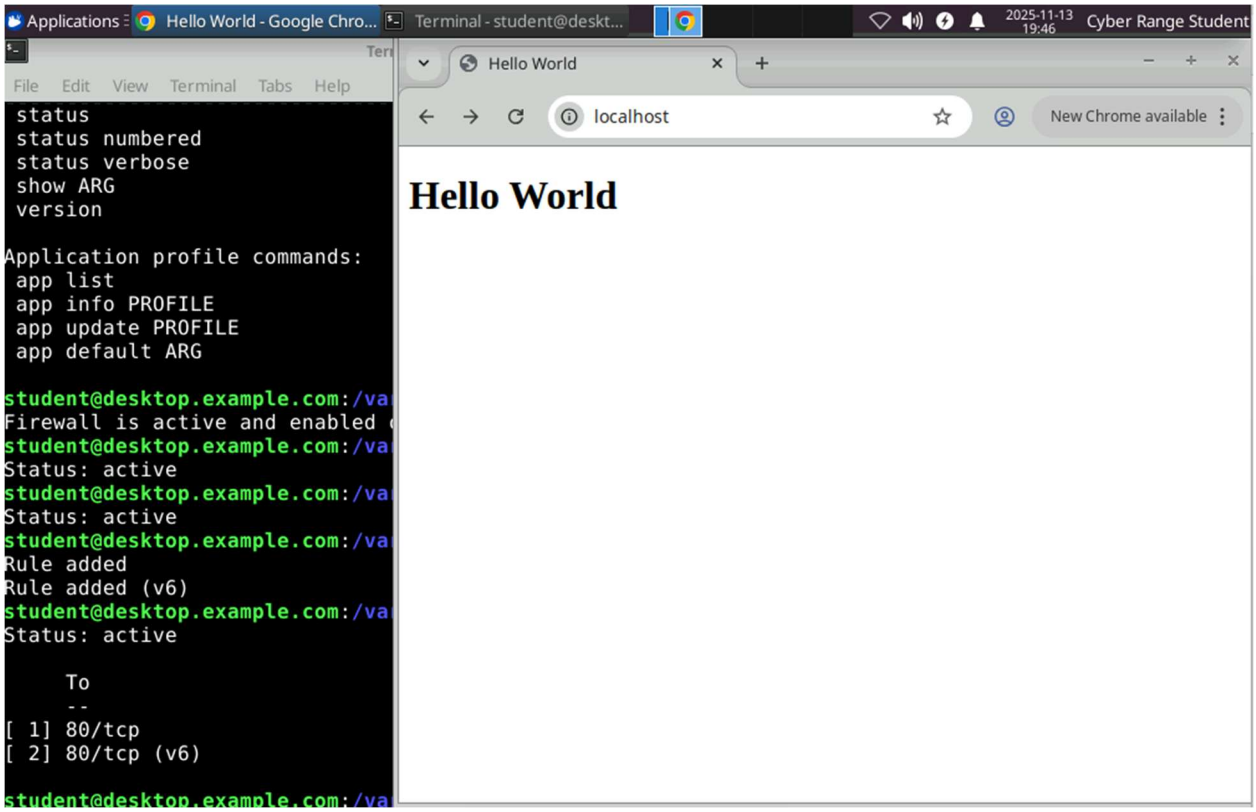
I verified firewall rules: sudo ufw status
If Apache was blocked, I would have run: sudo ufw allow 'Apache'
However, in the Cyber Range environment, the firewall typically allows internal HTTP traffic by default.

```
student@desktop.example.com:/var/www/html$ sudo ufw enable
Firewall is active and enabled on system startup
student@desktop.example.com:/var/www/html$ sudo ufw status
Status: active
student@desktop.example.com:/var/www/html$ sudo ufw status numbered
Status: active
student@desktop.example.com:/var/www/html$ sudo ufw allow http
Rule added
Rule added (v6)
student@desktop.example.com:/var/www/html$ sudo ufw status numbered
Status: active

     To                         Action      From
     --                         ------      ----
[ 1] 80/tcp                     ALLOW IN    Anywhere
[ 2] 80/tcp (v6)                ALLOW IN    Anywhere (v6)

student@desktop.example.com:/var/www/html$
```

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Reflection

This assignment showed me how something as simple as a port conflict or previous configuration can prevent an entire service from starting or operating properly. At first I assumed that the issue was file permissions, but testing with my new found command ss -tulpn revealed that nginx was occupying port 80.

Troubleshooting these problems helped me reinforce several real-world skills:

- How Linux services interact with each other
- How to diagnose service failures using systemctl status and logs
- How web servers bind to ports and why conflicts occur
- How document roots and index files determine what a user sees in their browser

Understanding these concepts is essential for system administrators, web server deployment, IT security, and DevOps. In real-world scenarios, multiple services often compete for the same resources, and being able to diagnose and resolve issues quickly is critical.